PHD COURSE · AALBORG UNIVERSITY · 2026

Bayesian Statistics for PhD Researchers

Course logistics and software setup

Lu Cheng and Ege Rubak · Department of Mathematical Sciences, Aalborg University

6 days · 10 groups · R + Stan in JupyterLab

9–17 November 2026 · 08:45–16:00

BEFORE WE START · 5 MINUTES

Tell us about your background

On your phone or laptop, go to

menti.com

and enter the code

[code]

We ask about

- your field and the data you work with
- your statistics background
- your programming experience, R, Python or other
- whether you have used Bayesian methods before

Anonymous. The answers help us choose examples and set the pace.

THE COURSE AT A GLANCE

Six days of learning by doing

6

course days

two lectures and two exercise sessions each day

10

fixed groups

6–7 students each (62 in total), mixed disciplines, for the whole course

5

ECTS

in-class work, short preparation, and a final assignment after the course

R

+ Stan

in JupyterLab notebooks, from Day 3 you write your own Stan models

The goal: understand Bayesian reasoning well enough to apply it, in Stan, to a problem from your own research.

DATES AND ROOMS · AALBORG UNIVERSITY

Six days in November, three buildings

Day	Date	Where	Topic
1	Mon 9 Nov	Pontoppidanstræde 101, room 1.001	Data, probability and uncertainty
2	Tue 10 Nov	Kroghstræde 3, room 4.110	Bayesian updating, one-parameter models
3	Thu 12 Nov	Pontoppidanstræde 101, room 1.001	Monte Carlo, MCMC and Stan
4	Fri 13 Nov	Kroghstræde 3, room 1.104	Joint and hierarchical models
5	Mon 16 Nov	Pontoppidanstræde 101, room 1.001	Regression and model checking
6	Tue 17 Nov	Niels Jernes Vej 8A, room 0.02	Logistic regression, the modelling workflow

Every day **08:45–16:00**. Extra group rooms: Day 2 4.130 and 4.131; Day 4 4.129 and 5.149; Day 6 0.01.

A COURSE DAY

08:45–16:00, no scheduled breaks

Time	Session	What happens
08:45–09:45	Lecture A	Follow-up on open questions, teaching, worked example
09:45–11:10	Exercise A	Exercises and your group’s question set; take a break when it suits you
11:10–11:55	Peer review A	Everyone back at 11:10: peer interviews, then plenary
11:55–12:30	Lunch	
12:30–13:30	Lecture B	As Lecture A
13:30–15:00	Exercise B	As Exercise A
15:00–15:45	Peer review B	Everyone back at 15:00: peer interviews, then plenary
15:45–16:00	Daily diary	3 things you learned, 3 things you are not yet familiar with

GROUPS AND PEER REVIEW

Every session ends with groups questioning each other

1 · During the exercises

Write a question set

Three questions with your own worked solutions: a concept, a computation and a pitfall.

2 · Peer interviews

Ask, then answer

Your fixed partner group (1↔2, 3↔4, …): two 10-minute rounds, odd groups ask first, then swap.

3 · Plenary

Two groups drawn at random

One interviews the other in front of the class; then each reporter asks the group’s hardest open question.

Questions go to a named person, and roles rotate: coder, checker, question writer, interviewer, reporter. Nothing is uploaded. Be ready: any group can be drawn.

GRADING

Half in class, half after the course

50 %

In-class activity

Each day the lecturer scores each **group** 0–4 on how actively it works. Everyone present gets the group's score.

50 %

Final assignment

Individual, after the course, scored out of 100: seven independent problems for everybody, with Stan models you write yourself.

70 %

To pass

Weighted total of at least 70 %, and attendance on at least 5 of the 6 days.

Effort is graded, not correctness. A wrong answer with real thinking, or a good question, counts in your favour.

ATTENDANCE

Attend at least 5 of the 6 days

If you must miss one day

1. **Tell the lecturer in advance**, before the day starts.
2. Within a week, submit your own solutions to that day's **in-class exercises**.
3. Also submit Parts A and B of that day's **extra exercises** (new problems, with hints).

The lecturer grades the make-up 0–4; it counts as your score for that day.

Not met if...

- you miss a day without telling the lecturer in advance
- you miss a day and submit no make-up
- you miss more than one day

Then the course cannot be passed.

FINAL ASSIGNMENT

Seven independent problems for everybody

FOR EVERYBODY

Everyday settings

Each problem has its own small dataset and uses no results from the others, so you can do them in any order. If you get stuck in one, move on to the next.

Individual; most code is provided.

THE SEVEN PROBLEMS

From free throws to a bakery

Free throws · help-desk calls · ice-cream sales · restaurant reviews · driving test · kitchen scale · bakery

Updating, Stan, regression, pooling, logistic models, a simulation check and a decision.

Scored out of 100 on a five-area rubric. You fill in the missing code; you may be asked to explain it.

Deadline: 13 December 2026, 23:55 on Moodle. Strict: late submissions are not accepted.

SOFTWARE ENVIRONMENT

Everything lives in one conda environment: bayes

MANAGER

Miniconda

installs and isolates everything below

LANGUAGE

R

data, plots and all course code

MODELS

CmdStan + CmdStanR

compiles Stan models to C++ and runs them from R

NOTEBOOKS

JupyterLab

with the **R (bayes)** kernel

Four steps, about 30–60 minutes, most of it downloading: **Miniconda → C++ tools → the bayes environment → JupyterLab.**

Not required before Day 1, but trying it at home saves class time. Full guide: **installations.md** in the course folder.

STEPS 1–2 · MINICONDA AND C++ TOOLS

Install conda, then the compiler tools

	1 · Miniconda	2 · C++ tools
macOS	Installer for your chip (arm64 or x86_64), run with bash in Terminal; answer yes.	In Terminal: xcode-select --install
Windows	The .exe installer: Just Me, default folder, not on PATH. Then use Anaconda Prompt.	Nothing: it comes with step 3.
Linux	Installer for x86_64 or aarch64, run with bash; answer yes.	sudo apt install build-essential (Fedora: gcc-c++ make)

✓ Check: close and reopen the terminal, then **conda --version** prints a version and the prompt starts with **(base)**.

STEP 3 · THE COURSE ENVIRONMENT

Create bayes with R, CmdStan and the tidyverse

```
# in Terminal (macOS/Linux) or Anaconda Prompt (Windows)
conda create -n bayes -c conda-forge r-base r-cmdstanr r-tidyverse
conda activate bayes
```

What you get

R, CmdStanR, CmdStan, the C++ toolchain and the tidyverse (ggplot2, dplyr), all inside **bayes**. A few GB; several minutes.

Check

The prompt starts with **(bayes)**; **R --version** works; **CMDSTAN** points inside the bayes environment.

Keep -c conda-forge. These packages must not come from Anaconda's defaults channel.

STEP 4 · JUPYTERLAB

Add JupyterLab and the R (bayes) kernel

```
# still in the activated bayes environment
conda install -c conda-forge jupyterlab r-irkernel
Rscript -e "IRkernel::installspec(name='ir-bayes',
displayname='R (bayes)', user=FALSE, prefix=Sys.getenv('CONDA_PREFIX'))"
jupyter kernelspec list
```

✓ Check: the list includes **ir-bayes**. In JupyterLab it is called **R (bayes)**. Run these in the terminal, not in R.

THE FINAL CHECK

Compile and run a Stan model

```
# in a notebook cell with the R (bayes) kernel
library(cmdstanr)
cmdstan_version()
file <- file.path(cmdstan_path(), "examples",
"bernoulli", "bernoulli.stan")
mod <- cmdstan_model(file)
fit <- mod$sample(data = list(N = 10,
y = c(0,1,0,0,0,0,0,0,0,1)), seed = 123)
fit$summary("theta")
```

Success looks like

A one-row summary for **theta** with a mean near **0.2**.

The first compile takes about a minute; that is normal.

Then **library(tidyverse)** loads without an error.

Start JupyterLab with **conda activate bayes**, then **cd** to the course folder, then **jupyter lab**.

PREPARATION · BEFORE THE COURSE

One notebook: pre-course-refresher.ipynb

Part	Topic
0	Install, then check that R and Stan work
1	R essentials with real data
2	Probability and Bayes' rule
3	A little calculus: densities, logs, maxima
4	Simulation and a first likelihood
5–6	Your research question; self-check

How to use it

Open it in JupyterLab with the **R (bayes)** kernel. Everything is inside: text, data, exercises, hints.

Try first, then check the solutions notebook beside it. Not graded.

No RStudio, no separate scripts: every exercise set in the course is a single notebook.

EVERY TIME YOU WORK

Activate first, then launch

```
conda activate bayes
cd "/path/to/course"
jupyter lab
# Windows: cd /d "C:\path\to\course"
# keep this terminal open; Ctrl+C to stop
```

If something goes wrong

Symptom	Fix
R code gives Python errors	Kernel › Change Kernel › R (bayes)
R (bayes) is missing	Rerun the kernel registration, restart JupyterLab
CmdStan not found	Quit JupyterLab; conda activate bayes; relaunch

Use **Kernel › Restart Kernel and Run All Cells** before you submit, to check your notebook runs from a clean start.

BEFORE DAY 1

Try the install at home; bring your laptop either way

Setup help on Day 1

The first exercise session reserves time for installation. If your laptop is stuck, work on a group partner's machine meanwhile.

Course materials

All materials are on the course Moodle page (phd.moodle.aau.dk, course 2862). Start with the pre-course refresher.

Stuck?

Send the first error message and your operating system to luch@math.aau.dk.

Stan compilation is not needed on Day 1, but you will need a working setup of your own by Day 3.